KRISH OSWAL

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EDUCATION

Vishwakarma University — Pune, MH
Bachelor of Technology in Computer Science and Engineering; CGPA: 7.2 — *2022 to 2027*
Coursework: Data Structures and Algorithms, Object-Oriented Programming, DBMS, Operating Systems, Computer Networks, Machine Learning

St. Vincent's High School and Junior College — Pune, MH
HSC, PCM with Computer Science; 75% — *2020 to 2022*

PROJECTS

GeoTrade: Supply-Chain Exposure Map | *Python, FastAPI, PostgreSQL, PostGIS, React, Three.js*

- Built a 3D globe that shows which companies get hit when a supplier, port or country runs into trouble, covering **23,020 public companies in 90 countries**. Final year project.
- Wrote the Python pipeline that turned **29,442 messy exchange listings** into one clean, geocoded company list by merging cross-listings and duplicate share classes.
- Designed a **25-table PostGIS schema** and a **FastAPI** backend that answers every drill-down in **under 60 ms**.

Serverless Movie Recommender on AWS | *Python, AWS Lambda, DynamoDB, S3, Anthropic API*

- Built a recommender that puts a movie the user actually liked in their top 10 for **68% of test users, up from 18%**, by mixing **TF-IDF** content similarity with a popularity score.
- Served it as a **serverless API on AWS** (API Gateway, Lambda, S3, DynamoDB) defined in AWS SAM with least-privilege IAM, answering warm requests in **under 1 ms** from a **19 KB** package.
- Wrote a **trie** for autocomplete (**about 500x faster** than scanning) and an **LRU cache** that hit **75%** on 100K replayed requests, plus plain-English search with the **Anthropic API** that only returns real movies.

Placement Portal: Campus Recruitment System | *Java 17, Spring Boot, JPA, MySQL, JUnit, Docker*

- Built a **Spring Boot** portal where placement officers, faculty and students run a full hiring season: company drives, eligibility checks and applications.
- Kept all eligibility rules in one service layer, backed by a database unique key, so **no student can apply to a drive they don't qualify for** or apply twice, from any screen or API.
- Set it up so every push runs **16 automated tests** and a **Docker** build on **GitHub Actions** before anything ships.

IPL Analytics and Win Predictor | *Python, Pandas, DuckDB, SQL, Scikit-learn, React*

- Tested a famous cricket belief on **1,243 IPL matches**: winning the toss gives no real edge (51.4%, p = 0.35), but batting second does (54.5%).
- Caught data leakage and **20 wrong match results** in my own first predictor, fixed them, and built a ball-by-ball model that calls chases right **83.6% of the time** on seasons it never saw.
- Cleaned 19 seasons of data with Python and **DuckDB SQL** and built an interactive dashboard with 21 charts.

TECHNICAL SKILLS

Languages: Java, Python, C++, SQL, TypeScript, JavaScript
Frameworks and Databases: Spring Boot, Spring Data JPA, FastAPI, React, PostgreSQL, PostGIS, MySQL, DuckDB
Data and ML: Pandas, NumPy, Scikit-learn, TensorFlow, Keras, hypothesis testing, TF-IDF
Cloud and Tools: AWS (Lambda, API Gateway, S3, DynamoDB), Docker, Git, GitHub Actions, OpenAI, Anthropic
Core Concepts: Data Structures and Algorithms, OOP, Design Patterns (MVC, Repository, DTO), REST API Design, Schema Design and Indexing, Unit and Integration Testing, CI/CD

CERTIFICATIONS

Python Programming: Tutedude — 2026
Data Structures and Algorithms in Python: Tutedude — 2025
SQL (Intermediate): HackerRank — 2025
Responsive Web Design: freeCodeCamp — 2025